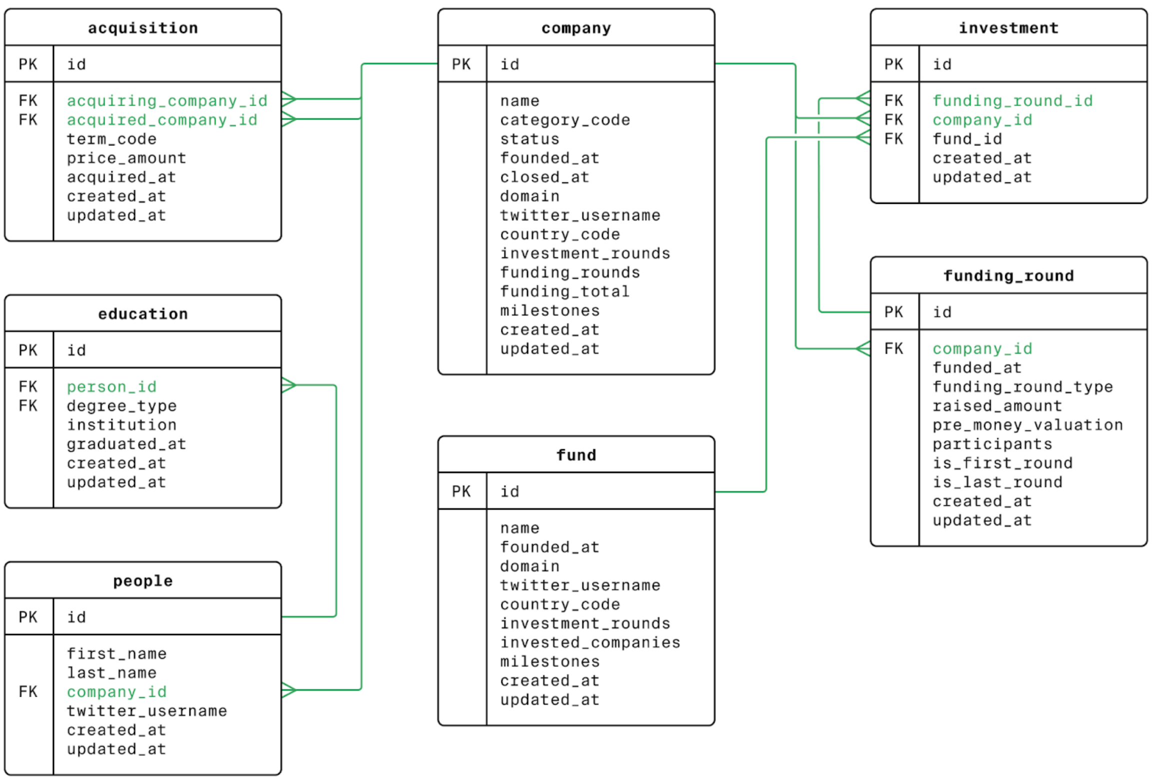
 Data Analyst at VentureInsight, a leading research firm that provides analytics and insights to venture capital firms and startup investors. Our clients rely on our data-driven recommendations to make multi-million dollar investment decisions.

We've just acquired a comprehensive database tracking venture funds, startups, acquisitions, and key people in the industry. Your manager has assigned you a series of analysis tasks that will help shape our upcoming quarterly investment report.

The database structure is shown in the following ER diagram:



1.

Startup Landscape Analysis

Before diving into specific analyses, your first task is to understand the overall startup landscape in our database. The executive team needs a snapshot of how many companies have failed (closed down) versus how many are still operating or have been acquired. This will help establish the baseline success rate in the startup ecosystem.

2.

Sector Analysis for US Investors

One of our major clients, a US-based VC firm, is considering investments in the media and news space. They've asked us to provide data on how much funding news-related companies from the USA have raised historically, to help them benchmark appropriate investment amounts.

```sql
SELECT funding_total
FROM company
WHERE category_code = 'news' AND country_code = 'USA'
ORDER BY funding_total DESC;
```

Result

funding_total
6.22553e+08
2.5e+08
1.605e+08
1.28e+08
1.265e+08
7e+07

3.

Analyzing Cash Acquisitions

Our quarterly report includes a section on acquisition trends. The team needs to understand the volume of cash-based acquisitions (as opposed to stock deals) that occurred during the recent post-recession period (2011-2013). This data will help identify whether companies were primarily acquired with cash or other payment methods during this economic recovery period.

```sql
SELECT SUM(price_amount)
FROM acquisition
WHERE term_code = 'cash'
  AND (acquired_at >= '2011-01-01' AND acquired_at <= '2013-12-31');
```

Result

sum
1.37762e+11

4.

Identifying Industry Influencers

Our marketing team is preparing an outreach campaign to industry influencers with strong social media presence. They're particularly interested in individuals who brand themselves with "Silver" in their Twitter handles, as this group seems to have significant industry clout. We need to identify these individuals for potential partnerships.

```sql
SELECT
    first_name,
    last_name,
    twitter_username
FROM people
WHERE twitter_username LIKE 'Silver%';
```

Result

first_name	last_name	twitter_username
Rebecca	Silver	SilverRebecca
Silver	Teede	SilverMatrixx
Mattias	Guilotte	Silverreven

5.

Finding Finance Influencers

Following your initial influencer analysis, the marketing team has refined their focus. They're now looking specifically for finance-focused influencers (those with "money" in their Twitter handles) whose last names start with 'K'. This more targeted approach will help them connect with relevant industry voices for our upcoming FinTech investment report.

```sql
SELECT *
FROM
    people
WHERE twitter_username LIKE '%money%' AND last_name LIKE 'K%'
```

Result

id	first_name	last_name	company_id	twitter_username	created_at	updated_at
63081	Gregory	Kim		gmoney75	2010-07-13 03:46:28	2011-12-12 22:01:34

6.

Geographic Investment Analysis

Our global investment clients need to understand funding patterns across different countries. They want to identify which countries attract the most venture capital to help them decide where to focus their international investment strategies. This geographic breakdown will be a key feature in our quarterly global trends report.

```sql
SELECT
    country_code,
    SUM (funding_total)
FROM company
GROUP BY country_code
ORDER BY SUM(funding_total) DESC;
```

Result

country_code	sum
USA	3.10588e+11
GBR	1.77056e+10
	1.08559e+10
CHN	1.06897e+10
CAN	9.86636e+09
IND	6.14141e+09

7.

Funding Round Volatility Analysis

Our risk analysis team is examining volatility in funding rounds. They're specifically interested in dates where there was significant variation between the smallest and largest rounds. This indicates days when both very small and very large companies were receiving funding, which could signal unusual market activity. They also want to exclude days where some companies received no funding at all, as that skews the analysis.

```sql
SELECT
    funded_at,
    MIN(raised_amount),
    MAX(raised_amount)
FROM funding_round
GROUP BY funded_at
HAVING
    MIN(raised_amount) <> MAX(raised_amount) AND MIN(raised_amount) <> 0;
```

Result

funded_at	min	max
2012-08-22	40000	7.5e+07
2010-07-25	3.27825e+06	9e+06
2002-03-01	2.84418e+06	8.95915e+06
2010-10-11	28000	2e+08
2007-01-18	5.5e+06	2.3e+07
2007-02-27	1.29e+06	3.6e+07

8.

Fund Activity Classification

For our investor clients, understanding the activity level of different venture funds helps them identify potential co-investment partners. Funds that invest in many companies are often seen as having broader networks, while those with fewer investments might have deeper industry expertise. We need to categorize funds by their activity level to help our clients find appropriate partners.

```sql
SELECT *,
       CASE
           WHEN invested_companies >= 100 THEN 'high_activity'
           WHEN invested_companies >= 20 THEN 'middle_activity'
           ELSE 'low_activity'
       END AS activity
FROM fund;
```

Result

id	name	founded_at	domain	twitter_username	country_code	investment_rounds	investees
13131						0	0
1	Greylock Partners	1965-01-01	greylock.com	greylockvc	USA	307	196
10	Mission Ventures	1996-01-01	missionventures.com		USA	58	33
100	Kapor Enterprises, Inc.		kei.com		USA	2	1

9.

Investment Strategy by Fund Activity

Building on our fund activity classification, our research team wants to understand how a fund's investment approach changes based on its activity level. Specifically, we want to know if funds that invest in more companies tend to participate in more funding rounds per company. This will help our clients understand different fund strategies and how broadly or deeply funds typically engage with their portfolio companies.

```sql
SELECT CASE
           WHEN invested_companies >= 100 THEN 'high_activity'
           WHEN invested_companies >= 20 THEN 'middle_activity'
           ELSE 'low_activity'
       END AS activity,
       ROUND(AVG(investment_rounds)) AS average_rounds
FROM fund
GROUP BY activity
ORDER BY average_rounds;
```

Result

activity	average_rounds
low_activity	2
middle_activity	51
high_activity	252

10.

Employee Education Impact on Startup Success

A heated debate has emerged among our clients about whether the educational background of startup employees correlates with company success. Some argue that highly educated teams are more likely to succeed, while others claim education has little impact. To settle this debate with data, we need to compare the education levels of employees at successful companies versus those that closed after limited funding.

```sql
SELECT AVG(t.total_degree_type)
FROM (SELECT p.id,
      COUNT(e.degree_type) AS total_degree_type
      FROM people AS p JOIN education AS e ON p.id = e.person_id
      WHERE company_id IN (SELECT id
                           FROM company
                           WHERE id IN (SELECT company_id
                                        FROM funding_round
                                        WHERE is_first_round = 1 AND is_last_round = 1)
                                        AND status = 'closed')
      GROUP BY p.id) AS t;
```

Result

avg
1.23113